

61. (a) Let $p(x) = x^3 - 3x^2 + kx - 10$

Then, $(x-5)$ is a factor of $p(x)$.

$$\begin{aligned}\therefore p(5) &= 0 \\ \Rightarrow 5^3 - 3(5)^2 + k \times 5 - 10 &= 0 \\ \Rightarrow 125 - 75 + 5k - 10 &= 0 \\ \Rightarrow 40 + 5k &= 0 \Rightarrow k = -8.\end{aligned}$$

62. (b) $\because x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)$

Given, $x + y + z = 0$

$$\begin{aligned}\therefore x^3 + y^3 + z^3 - 3xyz &= 0 \\ \Rightarrow x^3 + y^3 + z^3 + 3xyz &= 6xyz\end{aligned}$$

63. (c) $x^3q^2 - x^3pt + 4x^2p^2t - 4x^2q^2t + 3xq^2 - 3xpt$
 $= x^3q^2 - x^3pt - 4x^2q^2t + 4x^2p^2t + 3xq^2 - 3xpt$
 $= x^3(q^2 - pt) - 4x^2(q^2 - pt) + 3x(q^2 - pt)$
 $= (x^3 - 4x^2 + 3x)(q^2 - pt)$
 $= x(x^2 - 4x + 3)(q^2 - pt)$
 $= x(x^2 - 3x - x + 3)(q^2 - pt)$
 $= x[x(x-3) - 1(x-3)](q^2 - pt)$
 $= x(x-3)(x-1)(q^2 - pt)$

64. (c) In option (c), we have

$$\begin{aligned}(2)^4 - 6(2)^3 + 12(2)^2 - 24 \times 2 + 32 \\ = 16 - 6(8) + 12(4) - 48 + 32 \\ = 16 - 48 + 48 - 48 + 32 = 0\end{aligned}$$

Hence, $(x-2)$ is a factor of

$$x^4 - 6x^3 + 12x^2 - 24x + 32$$

65. (d) Let $f(x) = x^3 + kx^2 - x + 2$

If $(x+1)$ is a factor of $f(x)$.

Then, $f(-1) = 0$

$$\begin{aligned}\Rightarrow (-1)^3 + k(-1)^2 - (-1) + 2 &= 0 \\ \Rightarrow -1 + k + 1 + 2 &= 0 \Rightarrow k = -2\end{aligned}$$

66. (c) Let $f(x) = x^6 + px^5 + qx^4 - x^2 - x - 3$

$\because f(x)$ is divisible by $(x^4 - 1)$.

Then, $f(1) = f(-1) = 0$

Now, $f(1) = 1 + p + q - 1 - 1 - 3 = 0$

$$\Rightarrow p + q = 4 \quad \dots(i)$$

and $f(-1) = 1 - p + q - 1 + 1 - 3 = 0$

$$\Rightarrow q - p = 2 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get $q = 3$ and $p = 1$

$$\therefore p^2 + q^2 = 3^2 + 1^2 = 10$$

67. (a) Let $f(x) = ax^2 - (a^2 + 1)x + a$
 $= a x^2 - x - a^2 x + a$

$$= x(ax - 1) - a(ax - 1)$$

$$= (x - a)(ax - 1)$$

If p and q are linear factors of $f(x)$, then

$$p = (x - a) \text{ and } q = (ax - 1)$$

$$\begin{aligned}\therefore p + q &= (x - a) + (ax - 1) \\ &= (x - 1)(a + 1)\end{aligned}$$

68. (d) We have, $x - y = 8 \quad \dots(i)$

$$\text{and } \sqrt{\frac{x}{y}} = \frac{10}{3} - \sqrt{\frac{y}{x}}$$

$$\Rightarrow \sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = \frac{10}{3}$$

$$\Rightarrow \frac{(\sqrt{x})^2 + (\sqrt{y})^2}{\sqrt{xy}} = \frac{10}{3}$$

$$\Rightarrow x + y = \frac{10}{3}\sqrt{xy}$$

On squaring both sides, we have

$$(x + y)^2 = \frac{100}{9}xy$$

$$\Rightarrow (x - y)^2 + 4xy = \frac{100}{9}xy$$

$$\begin{aligned}[\because (x + y)^2 &= (x - y)^2 + 4xy] \\ \Rightarrow (8)^2 &= \left(\frac{100}{9} - 4\right)xy \quad [\text{from Eq. (i)}]\end{aligned}$$

$$\Rightarrow 64 = \frac{64}{9}xy \Rightarrow xy = 9$$

HCF AND LCM OF POLYNOMIALS

Regularly (1-2) questions have been asked from this chapter. Generally questions are based on your prior knowledge of GCD and LCM of numbers and expressions.



DIVISOR

A polynomial $d(x)$ is said to be a divisor of polynomial $p(x)$ if $d(x)$ is a factor of $p(x)$ i.e., $p(x)$ can be written as $p(x) = d(x) \cdot q(x)$, where $q(x)$ is a quotient polynomial e.g. $(x-3)$ is a divisor of $(x-3)^2(x+1)$.

HCF (GCD) of Polynomials

A polynomial $h(x)$ is called the HCF or GCD of two or more given polynomials, if $h(x)$ is a polynomial of highest degree dividing each one of the given polynomial without leaving any remainder.

Note The coefficient of highest degree term is always taken as positive.

EXAMPLE 1. The HCF of polynomials $(x+2)^2(x-3)^2$ and $(x+1)(x+2)^2(x-3)$ is

a. $(x+2)(x-3)$

b. $(x+2)^2(x-3)$

c. $(x+1)(x+2)(x-3)$

d. $(x+2)^2(x-3)^2$

Sol. b. Let $p(x) = (x+2)^2(x-3)^2$ and $q(x) = (x+1)(x+2)^2(x-3)$

The common factor between two polynomials is $(x+2)^2(x-3)$, $\therefore$ HCF = $(x+2)^2(x-3)$

HCF by Factorisation Method

Following are the steps for calculating HCF through factorisation method

- Step I** Resolve the given polynomials in the complete factored form.
- Step II** Find the HCF of the numerical factors (if any) of given polynomial.
- Step III** Find the factors of highest degree which is common to all given polynomials.
- Step IV** The product of all such common factors and HCF of the numerical factors is the HCF of given polynomials.

EXAMPLE 2. The HCF of $p(x) = 24(6x^4 - x^3 - 2x^2)$ and $q(x) = 20(2x^6 + 3x^5 + x^4)$ is

- a. $4(2x+1)$ b. $x^2(2x+1)$ c. $4x^2$ d. None of these

Sol. d. Here, $P(x) = 24(6x^4 - x^3 - 2x^2) = 2^3 \cdot 3 \cdot x^2 \cdot (6x^2 - x - 2)$
 $= 2^3 \cdot 3 \cdot x^2 \cdot (6x^2 - 4x + 3x - 2) = 2^3 \cdot 3 \cdot x^2(2x+1) \cdot (3x-2)$
 and $q(x) = 20(2x^6 + 3x^5 + x^4) = 2^2 \cdot 5 \cdot x^4(2x^2 + 3x + 1)$
 $= 2^2 \cdot 5 \cdot x^4 \cdot (2x^2 + 2x + x + 1) = 2^2 \cdot 5 \cdot x^4(2x+1)(x+1)$
 HCF of numerical factor $= 2^2 = 4$, Highest degree common factor $= x^2(2x+1)$
 $\therefore$ Required HCF $= 4x^2(2x+1)$

HCF by Division Method

To find the HCF of polynomials which cannot be factorised easily, we use successive division method.

- Step I** Arrange the given polynomials in descending order of powers of its variables.
Step II If any common factor is present in the terms of each polynomial, it should be taken out.
Step III Divide the polynomial of highest degree by the polynomial of lowest degree.
 or If both the polynomials are of the same degree then any one of them can be taken as divisor or dividend.
Step IV After the first division take the remainder as the new divisor and first divisor as new dividend.
Step V Continue this process of dividing the last divisor by the last remainder until the remainder becomes zero.
Step VI The product of common factors obtained from step II and last divisor is the HCF of given polynomials.

- Note**
- If the first term of a remainder is negative at any stage, the sign of all of its term must be changed.
 - If at any stage, the remainder contains common factor it should be taken out.

EXAMPLE 3. The HCF of $22x^6 - 78x^5 - 16x^2$ and $2x^5 - 78x^2 - 44x$ is

- a. $(x^2 - 3x - 2)$ b. $2x(x^2 - 3x - 2)$
 c. $22x(x^2 - 3x - 2)$ d. None of these

Sol. b. Let $p(x) = 22x^6 - 78x^5 - 16x^2 = 2x^2(11x^4 - 39x^3 - 8)$
 $q(x) = 2x^5 - 78x^2 - 44x = 2x(x^4 - 39x - 22)$
 Let us divide $(11x^4 - 39x^3 - 8)$ by $(x^4 - 39x - 22)$

$$\begin{array}{r} x^4 - 39x - 22 \overline{) 11x^4 - 39x^3 - 8} \\ \underline{11x^4 - 429x - 242} \\ -39x^3 + 429x + 234 \end{array}$$

-39 is taken out as common factor from remainder.

$$\begin{array}{r} x^3 - 11x - 6 \overline{) x^4 - 11x^2 - 6x} \\ \underline{-x^3 + 11x + 6} \\ 11x^2 - 33x - 22 \end{array}$$

Again 11 is taken out as common factor from remainder

$$\begin{array}{r} x^2 - 3x - 2 \overline{) x^3 - 11x - 6} \\ \underline{x^3 - 3x^2 - 2x} \\ 3x^2 - 9x - 6 \\ \underline{3x^2 - 9x - 6} \\ 0 \end{array}$$

$\therefore$ HCF of $11x^4 - 39x^3 - 8$

and $x^4 - 39x - 22$ is $x^2 - 3x - 2$.

Also, HCF of $2x^2$ and $2x$ is $2x$.

$\therefore$ Required HCF $= 2x(x^2 - 3x - 2)$

LCM of Polynomials

A polynomial $b(x)$ is called the LCM of two or more polynomials, if it is a polynomial of smallest degree which is divided by each one of the given polynomials without leaving any remainder.

EXAMPLE 4. The LCM of $12x^2y^3z^2$ and $18x^4y^2z^3$ is

- a. $x^4y^3z^3$
 b. $9x^2y^2z^3$
 c. $36x^3y^3z^4$
 d. $36x^4y^3z^3$

Sol. d. Here, $12x^2y^3z^2 = 2^2 \times 3^1 \times x^2 \times y^3 \times z^2$
 $18x^4y^2z^3 = 2^1 \times 3^2 \times x^4 \times y^2 \times z^3$
 $\therefore$ Required LCM $= 2^2 \times 3^2 \times x^4 \times y^3 \times z^3$
 $= 36x^4y^3z^3$

LCM by Factorisation Method

Following are the steps for calculating LCM through factorisation method

- Step I** Resolve the given polynomials in the complete factored form
Step II Find the LCM of the numerical factors (if any) of given polynomials
Step III The required LCM is the product of LCM of numerical factors and each factor raised to the highest power.

EXAMPLE 5. The LCM of $x^3 - 2x^2 - x + 2$ and $x^3 - x^2 - 4x + 4$ is

- a. $(x^2 - 1)(x^2 - 4)$ b. $(x^2 + 2)(x^2 - 4)$
 c. $(x^2 + 1)(x^2 + 2)$ d. None of these

Sol. a. Here, $x^3 - 2x^2 - x + 2 = x^2(x - 2) - 1(x - 2)$

$$\Rightarrow (x - 2)(x^2 - 1) = (x - 2)(x - 1)(x + 1)$$

$$= (x - 1)(x + 1)(x - 2)$$

$$\text{and } x^3 - x^2 - 4x + 4 = x^2(x - 1) - 4(x - 1)$$

$$\Rightarrow (x - 1)(x^2 - 4) = (x - 1)(x - 2)(x + 2)$$

$$\therefore \text{ Required LCM} = (x - 1)(x + 1)(x - 2)(x + 2)$$

$$= (x^2 - 1)(x^2 - 2^2)$$

$$[\because (a - b)(a + b) = a^2 - b^2]$$

$$= (x^2 - 1)(x^2 - 4)$$

For any two polynomials $p(x)$ and $q(x)$

$$p(x) \times q(x) = (\text{Their HCF}) \times (\text{Their LCM})$$

i.e. Product of two polynomials

= Product of their HCF and LCM

EXAMPLE 6. The HCF of two polynomials is $x + 3$ and their LCM is $x^3 - 7x + 6$. If one of the polynomials is $x^2 + 2x - 3$. Then, the other is

- a. $x^2 - x - 6$ b. $x^2 - x + 6$
 c. $x^2 + x + 6$ d. None of these

Sol. d. Here, HCF = $x + 3$ and LCM = $x^3 - 7x + 6$

Also, one polynomial = $x^2 + 2x - 3$

As, LCM $\times$ HCF = Product of two polynomials

$$\text{Other polynomial} = \frac{\text{LCM} \times \text{HCF}}{\text{One polynomial}}$$

$$= \frac{(x + 3) \times (x^3 - 7x + 6)}{x^2 + 2x - 3}$$

$$= \frac{(x + 3)(x^3 - 7x + 6)}{(x + 3)(x - 1)}$$

$$= \frac{(x + 3)(x - 1)(x^2 + x - 6)}{(x + 3)(x - 1)} = x^2 + x - 6$$

So, other polynomial is $x^2 + x - 6$.

PRACTICE EXERCISE

- The LCM of $(x - 1)(x - 2)$ and $x^2(x - 2)(x + 3)$ is
 (a) $(x - 1)$ (b) $(x - 1)(x - 2)(x + 3)$
 (c) $x^2(x - 1)(x - 2)(x + 3)$ (d) None of these
- The LCM of $2(a^2 - b^2)$, $3(a^3 - b^3)$, $4(a^4 - b^4)$ is
 (a) $6(a - b)(a + b)(a^2 + b^2)$ (b) $12(a^4 - b^4)(a^2 + ab + b^2)$
 (c) $a^3 - b^3$ (d) $12(a^4 - b^4)$
- The HCF of two expressions a and b is 1. Their LCM is
 (a) $(a + b)$ (b) $a - b$ (c) ab (d) $\frac{1}{ab}$
- The LCM of the polynomials $(x + 3)^2(x - 2)(x + 1)^2$; $(x + 1)^3(x + 3)(x + 4)$ is
 (a) $(x - 2)(x + 1)^3(x + 3)^2(x + 4)$
 (b) $(x - 2)(x + 1)^3(x + 3)(x + 4)$
 (c) $(x - 2)(x + 3)(x + 4)$
 (d) $(x - 2)^2(x + 1)(x + 3)^2(x + 4)$
- HCF of $4y^4x - 9y^2x^3$ and $4y^2x^2 + 6yx^3$ is
 (a) $y^2x(2y + 3x)$ (b) $yx(3x + 2y)$
 (c) $yx^2(x + 3)$ (d) None of these
- What is the HCF of $(x^4 - x^2 - 6)$ and $(x^4 - 4x^2 + 3)$?
 (a) $x^2 - 3$ (b) $x + 2$ (c) $x + 3$ (d) $x^2 + 3$
- The HCF of the polynomial A and B where
 $A = (x + 3)^2(x - 2)(x + 1)^2$
 and $B = (x + 1)^2(x + 3)(x + 4)$ is given by
 (a) $(x + 1)^2(x + 3)$ (b) $(x + 1)(x + 3)^2$
 (c) $(x + 1)(x + 3)$ (d) $(x + 3)^2(x + 1)^2$
- The HCF of $22x(x + 1)^2$ and $36x^2(2x^2 + 3x + 1)$ is
 (a) $2x(x + 1)$ (b) $x(x + 1)$ (c) $2(x + 1)$ (d) $2(x + 1)^2$
- The LCM of $a^2 - b^2 - c^2 - 2bc$,
 $b^2 - c^2 - a^2 - 2ac$ and $c^2 - a^2 - b^2 - 2ab$ is
 (a) $(a + b + c)$
 (b) $(a - b - c)(a + b + c)$
 (c) $(a + b + c)(c - a - b)$
 (d) $(a + b + c)(a - b - c)(b - c - a)(c - a - b)$
- LCM of $[(x + 3)(x - 2)]^2$ and $[(x - 2)(x - 6)]$ is
 (a) $(x + 3)(x - 2)^3(x - 6)$ (b) $(x + 3)(x - 2)^2(x - 6)$
 (c) $(x + 3)(x - 2)(x - 6)$ (d) $(x + 3)(x - 6)$

- 11.** If $(z - 1)$ is the HCF of $(z^2 - 1)$ and $pz^2 - q(z + 1)$, then
 (a) $2p = q$ (b) $p = 2q$ (c) $3p = 2q$ (d) $3p = 2p$
- 12.** The HCF of two expressions is $3x^2 + 4x - 4$ and their LCM is $3x^4 + 4x^3 - 7x^2 - 4x + 4$. The expressions are
 (a) $(x - 1)(3x^2 + 4x - 4)$ and $(3x^2 + 4x - 4)$
 (b) $(x + 1)(3x^2 + 4x - 4)$ and $(x + 2)(3x^2 + 4x - 4)$
 (c) $(x + 2)(3x^2 + 4x + 4)$ and $(x - 1)$
 (d) $(x + 1)(3x^2 + 4x - 4)$ and $(x - 1)(3x^2 + 4x - 4)$
- 13.** The LCM and HCF of two polynomials $p(x)$ and $q(x)$ are $36x^2(x + a)(x^3 - a^3)$ and $x^2(x - a)$, respectively. If $p(x) = (4x^2)(x^2 - a^2)$, then $q(x)$
 (a) $4x^3(x^3 - a^3)$ (b) $12x^3(x^2 - a^2)$
 (c) $9x^3(x^3 - a^3)$ (d) $36x^3(x^3 - a^3)$
- 14.** The LCM of two polynomials $p(x)$ and $q(x)$ is $x^3 - 7x + 6$. If $p(x) = x^2 + 2x - 3$ and $q(x) = x^2 + x - 6$, then the HCF is
 (a) $(x + 3)$ (b) $(x - 3)$
 (c) $(x + 3)(x - 2)$ (d) $(x - 1)$
- 15.** If $(x + k)$ is the HCF of $(x^2 + ax + b)$ and $(x^2 + px + q)$, then the value of k is
 (a) $\left(\frac{b+q}{a+p}\right)$ (b) $\left(\frac{b-q}{a-p}\right)$ (c) $\left(\frac{a+b}{p+q}\right)$ (d) $\left(\frac{a-b}{p-q}\right)$
- 16.** What is the value of k for which the HCF of $2x^2 + kx - 12$ and $x^2 + x - 2k - 2$ is $(x + 4)$?
 (a) 5 (b) 7 (c) 10 (d) -4
- 17.** If the HCF of $(x^2 + x - 12)$ and $(2x^2 - kx - 9)$ is $(x - k)$, then what is the value of k ?
 (a) -3 (b) 3 (c) -4 (d) 4
- 18.** If GCD of the polynomials $(x^3 - 2x^2 + px + 6)$ and $(x^2 - 5x + q)$ is $(x - 3)$. Then, the value of $5q + 6p$ is
 (a) -1 (b) 1
 (c) 0 (d) None of these
- 19.** The sum and the difference of two expressions is $5x^2 - x - 4$ and $x^2 + 9x - 10$ respectively, then their LCM would be equal to
 (a) $(x - 1)$ (b) $(2x + 3)(3x + 7)$
 (c) $(2x - 3)(3x + 7)$ (d) $(x - 1)(2x - 3)(3x + 7)$
- 20.** Find the values of a and b so that the polynomials $p(x)$ and $q(x)$ have $(x + 1)(x + 3)$ as their HCF

$$p(x) = (x^2 + 3x + 2)(x^2 + 2x + a)$$
 and $q(x) = (x^2 + 7x + 12)(x^2 + 7x + b)$
 (a) -3, 6 (b) 3, -6
 (c) 6, -3 (d) None of these
- 21.** If $f(x)$ and $g(x)$ are two polynomials with integral coefficients which vanish at $x = 1/2$, then what is the factor of HCF of $f(x)$ and $g(x)$?
 (a) $x - 1$ (b) $x - 2$ (c) $2x - 1$ (d) $2x + 1$
- 22.** The HCF of two polynomials $p(x)$ and $q(x)$ is $2x(x + 2)$ and LCM is $24x(x + 2)^2(x - 2)$. If $p(x) = 8x^3 + 32x^2 + 32x$, then what is $q(x)$ equal to?
 (a) $4x^3 - 16x$ (b) $6x^3 - 24x$ (c) $12x^3 + 24x$ (d) $12x^3 - 24x$
- 23.** If $(x - 2)$ is the HCF of $(ax^2 + bx + c)$ and $(bx^2 + ax + c)$, then value of c is
 (a) $2(a + b)$ (b) $(a + b)$ (c) $-3(a + b)$ (d) $-(a + b)$
- 24.** We have three polynomials $A = 8p + p^2 + 12$, $B = p^2 + 2p - 24$ and $C = p^2 + 15p + 54$
 I. Their LCM is $(p + 6)(p - 4)(p + 2)(p + 9)$
 II. Their HCF is $(p + 6)(p - 2)$
 Then, which of the following codes is/are correct
 (a) Only I (b) Only II
 (c) Neither I nor II (d) Both I and II
- 25.** Which of the following statements are true?
 I. HCF of $x^2 - 6x + 9$ and $x^3 - 27$ is $(x - 3)$.
 II. LCM of $10x^2yz$, $15xyz$, $20xy^2z^2$ is $120x^2y^2z^2$.
 III. HCF of $(6x^2 - 7x - 3)$ and $(2x^2 + 11x - 21)$ is $(2x - 3)$.
 Select the correct answer using the codes given below
 (a) I and III (b) I, II and III
 (c) II and III (d) None of these
- 26.** Consider the following statements :
 I. The HCF of $x + y$ and $x^{10} - y^{10}$ is $x + y$.
 II. The HCF of $x + y$ and $x^{10} + y^{10}$ is $x + y$.
 III. The HCF of $x - y$ and $x^{10} + y^{10}$ is $x - y$.
 IV. The HCF of $x - y$ and $x^{10} - y^{10}$ is $x - y$.
 Which of the statement(s) given above is/are correct?
 (a) I and II (b) II and III (c) I and IV (d) II and IV
- Directions** (Q. Nos. 27-29) A student wrote five polynomials such as $A = pq - np$, $B = pq - mq$, $C = q^2 - 3nq + 2n^2$, $D = pq - 2pn - mq + 2mn$, $E = pq - np - mq + mn$. Now, he divide the polynomials into groups and calculate the HCF and LCM.
- 27.** Calculate the HCF of A , C and E
 (a) $(q - 2n)$ (b) $(p - n)$ (c) $(q - n)$ (d) $(q - n)(q - 2n)$
- 28.** Calculate the LCM of D and E
 (a) $(p - m)(q - 2n)$ (b) $(p - m)(q - n)(q - 2n)$
 (c) $(q - n)(q - 2n)(m - p)$ (d) $(q - n)(p - m)$
- 29.** The HCF of all five polynomials together is
 (a) $(q - n)$ (b) $(q - 2n)$ (c) 1 (d) $(p - m)$